

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2- COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
Student Name:	SANTHOSH.K	Reg.No.:	192312450
Department:	ECE	Date:	26.8.26

ANNEXURE A
COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

I Santhosh K (192312450) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Santhosh K
Signature of the student:

MARKS ALLOCATION

	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)	
Q1	1	1.25	0.5	1	0.75	4.5
Q2	1	1	0.5	1	0.75	4.25
Q3	1	1	1	1	0.5	4.5
Q4	0.5	1.25	1	0.5	0.5	3.75
Q5	0.5	0.5	0.5	0.5	0.5	2.5
						19.5

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.

Compare L1, L2 and L3 Cache Levels

1. Scope of Items

The comparison covers the three major cache levels in a processor memory hierarchy. The important dimensions are access latency, bandwidth, effective throughput, capacity, physical location, sharing, cache-hit behavior and the resulting effect on processor performance. The assessment specifically asks for latency, bandwidth and throughput and their impact on processor performance.

2. Detailed Comparison

Parameter	L1 Cache	L2 Cache	L3 Cache
Relative size	Smallest	Medium	Largest cache level
Relative latency	Lowest	Low	Higher than L1/L2
Bandwidth	Highest effective local bandwidth	High	High but higher access delay
Typical location	Inside/very close to each core	Close to each core	Often shared among cores
Main purpose	Immediate working data/instructions	Catch L1 misses	Reduce DRAM accesses
Sharing	Usually private	Usually private or core-associated	Often shared
Hit benefit	Maximum	High	High compared with DRAM
Miss consequence	Access L2	Access L3/DRAM	Access DRAM
Area per bit	High	High	High
Power consideration	Important due to speed/size	Moderate/high	Higher total capacity can increase power
Best workload characteristic	Very high locality	Moderate/high locality	Shared/reused data

3. L1 Cache – Detailed Role

L1 is the first cache checked after the processor's registers and internal execution resources. Because its physical distance and size are small, it can be accessed with very low latency. Many processors separate L1 into instruction and data portions so that instruction fetching and data access can occur efficiently. The major limitation is capacity: only a small active working set can be retained.

4. L2 Cache – Detailed Role

L2 acts as an intermediate cache. It is larger than L1, so it can hold more of the working set, but its access latency is higher. An efficient L2 reduces the frequency with which L1 misses reach L3 or DRAM. L2 therefore provides an important balance between capacity and access speed.

5. L3 Cache – Detailed Role

L3 is generally larger and may be shared by multiple processor cores. A shared cache can reduce duplication of commonly used data and allow data produced by one core to be reused by another. However, because of its larger size and shared organization, its access latency is higher than L1 and L2.

6. Bandwidth and Throughput Interpretation

Bandwidth indicates how much data can be transferred per unit time, while throughput represents the amount of useful processing completed per unit time. A cache does not simply increase raw processor frequency; it improves effective throughput by reducing the time the processor spends waiting for memory. A high cache-hit rate means more requests are completed within fast cache levels.

7. Average Memory Access Time Concept

A useful performance concept is Average Memory Access Time (AMAT). A simplified expression is: $AMAT = \text{Hit Time} + \text{Miss Rate} \times \text{Miss Penalty}$. A well-designed hierarchy tries to keep both hit time and the combined miss penalty small. L1 provides very low hit time, while L2 and L3 reduce the probability that a miss reaches slower DRAM.

8. Critical Thinking

Increasing cache capacity can reduce miss rate, but it can also increase area, leakage power and access delay. Therefore, the design must optimize the workload rather than maximize cache size. For highly reusable data, additional cache capacity may be beneficial. For sequential streaming workloads with little data reuse, the additional cache may provide less improvement.

9. Practical Performance Evaluation

- Measure L1/L2/L3 cache hit and miss rates.
- Measure processor stall cycles caused by memory operations.
- Measure effective memory bandwidth and application throughput.
- Compare execution time with different cache configurations in a simulator or benchmark environment.
- Evaluate power and area trade-offs when cache size is increased.

10. Conclusion

L1, L2 and L3 provide progressively larger capacity at progressively higher access latency. L1 minimizes immediate access delay, L2 reduces L1 miss penalties, and L3 reduces expensive DRAM traffic. The hierarchy improves processor throughput by exploiting locality and keeping frequently reused data close to the CPU.

2.Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.

Differentiate SRAM and DRAM

1. Scope of Items

The comparison evaluates SRAM and DRAM according to latency, bandwidth and throughput and explains why SRAM is suitable for cache memory while DRAM is suitable for main memory.

2. Detailed Comparison

Parameter	SRAM	DRAM
Storage mechanism	Bistable/flip-flop circuit	Capacitor-based charge storage
Refresh	Not required	Required periodically
Access latency	Very low	Higher
Density	Lower	Higher
Capacity	Small/limited	Large
Cost per bit	Higher	Lower
Typical role	Cache/scratchpad	Main memory
Bandwidth	Very high for local access	High with wide interfaces/bursts
Power	No refresh, but larger cell	Refresh consumes power

	area/leakage	
Volatility	Volatile	Volatile
Design advantage	Speed	Density and capacity
Design limitation	Area/cost	Latency and refresh

3. SRAM Working Principle

SRAM stores a bit using a stable electronic state. As long as power is supplied, the stored state remains stable without periodic refresh. This makes SRAM well suited to high-speed cache structures. Its drawback is that the cell requires more circuitry than a DRAM cell, reducing density and increasing cost per stored bit.

4. DRAM Working Principle

DRAM stores a bit as electrical charge in a capacitor. Charge leakage means the information must be refreshed periodically. The compact memory cell allows a much larger amount of data to be stored in a relatively small physical area, making DRAM suitable for main memory.

5. Latency Comparison

SRAM has lower access latency because the stored state can be accessed directly through its circuitry without the refresh-related behavior of DRAM. DRAM access involves row activation and other memory operations and therefore has higher effective latency. Nevertheless, modern DRAM systems use wide buses, banks and burst transfers to provide high bandwidth.

6. Bandwidth and Throughput

SRAM is excellent for small, frequent, latency-sensitive accesses. DRAM is optimized to provide large amounts of data to the processor or accelerator. System throughput therefore depends on matching the technology to the workload. A processor benefits when its most frequently used data stays in SRAM caches while DRAM provides the capacity required by applications.

7. Cache and Main Memory Design

The natural hierarchical arrangement is CPU → SRAM-based cache → DRAM main memory. This design prevents the expensive requirement of building the entire main memory from SRAM while still providing very fast access to frequently used information.

8. Critical Thinking

If a system used only SRAM, it would have excellent latency but poor density and high implementation cost. If it used only DRAM, it would have adequate capacity but much higher access latency for the CPU. Therefore, the best architecture combines both technologies according to their strengths.

9. Evidence and Validation

- Compare cache hit time and DRAM access time using processor counters or simulation.
- Measure memory bandwidth under sequential and random access patterns.
- Measure application execution time before and after cache-size changes.
- Evaluate memory power, area and capacity constraints.
- Use workloads with different locality patterns to determine the benefit of SRAM cache.

10. Conclusion

SRAM is preferred for cache because of its low latency, while DRAM is preferred for main memory because of its high density and capacity. Their combination provides a balanced memory system with low effective access latency and practical storage capacity.

3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.

Compare Virtual Memory with Paging and Cache Memory

1. Scope of Items

The comparison focuses on access latency, throughput, purpose, management, data unit, hit/miss behavior and the effect of storage access. The assessment specifically asks for virtual memory with paging versus cache mechanisms. □filecite□turn1file0□L68-L68□

2. Detailed Comparison

Parameter	Cache Memory	Virtual Memory with Paging
Main purpose	Reduce memory access latency	Manage virtual address space
Data unit	Cache line/block	Page
Primary controller	Hardware/cache controller	OS + MMU/TLB
Fast case	Cache hit	Page resident in DRAM

Slow case	Cache miss	Page fault
Typical penalty	Lower memory level access	Potential storage I/O
Effect on CPU	Reduces stalls	Provides isolation/flexibility
Storage use	Normally not directly required	May be required after page fault
Locality principle	Strongly exploits locality	Uses page-level locality
Main risk	High miss rate	Thrashing/page faults

3. Cache Memory Mechanism

When the processor requests an address, the cache checks whether the corresponding cache line is present. If it is present, the processor obtains the data quickly. If it is absent, the system obtains the line from a lower memory level. The objective is to keep frequently reused data close to the processor.

4. Virtual Memory and Paging Mechanism

Virtual memory allows each process to use a virtual address space independent of the exact physical location of its data. Memory is divided into pages and physical memory into frames. The operating system maintains page mappings, while the MMU performs address translation. A TLB stores recently used translations to reduce translation overhead.

5. Latency Difference

A cache miss is normally handled by accessing another memory level, such as L2, L3 or DRAM. A page fault is much more expensive because the required page may need to be obtained from secondary storage. Therefore, cache misses and page faults should not be treated as equivalent events.

6. Throughput Difference

Caches increase processor throughput by reducing memory stalls. Virtual memory improves memory utilization, protection and flexibility. However, if a workload causes frequent page faults, the processor may spend substantial time waiting for storage I/O, reducing throughput. Excessive paging can lead to thrashing.

7. Critical Thinking

Cache and virtual memory are complementary rather than competing technologies. Cache is primarily a performance mechanism, while virtual memory is primarily an address-space management and protection mechanism. A well-designed system uses both and tries to keep active pages in DRAM while also keeping frequently accessed data in cache.

8. Evidence and Validation

- Measure cache miss rate and CPU stall cycles.
- Measure TLB hit/miss behavior where available.
- Measure page-fault rate and swap activity.
- Measure storage I/O generated by paging.
- Compare application throughput under adequate and insufficient physical-memory conditions.

9. Conclusion

Cache memory reduces processor access latency, whereas virtual memory with paging provides flexible memory management and process isolation. Cache misses are normally much less costly than page faults. Therefore, high cache locality and sufficient physical memory are both important for high system throughput.

4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.

NAND Flash versus DRAM

1. Scope of Items

The comparison evaluates NAND Flash and DRAM according to latency, bandwidth and suitability within a hierarchical memory system. ☐ filecite ☐ turn1file0 ☐ L69-L69 ☐

2. Detailed Comparison

Parameter	NAND Flash	DRAM
Data retention	Non-volatile	Volatile
Access latency	High	Low relative to Flash
Capacity	Very high	High
Bandwidth	Strong sequential throughput	High memory bandwidth
Random access	Slower	Much faster
Refresh	Not required	Required
Write behavior	Program/erase operations	Direct volatile memory writes
Endurance	Limited program/erase cycles	Different failure mechanism; not Flash-like
Typical application	SSD/embedded storage	Main memory
Hierarchy position	Persistent lower level	Active working memory

3. NAND Flash Characteristics

NAND Flash is a non-volatile semiconductor memory. It retains information after power is removed and therefore is ideal for persistent storage. Flash storage achieves high capacity through dense memory cells and parallel organization, but its program/erase operations and controller management make it much slower than DRAM for active memory access.

4. DRAM Characteristics

DRAM is volatile main memory. It provides much lower access latency than NAND storage and is designed for repeated read/write operations during program execution. Its major disadvantages are volatility and refresh requirements.

5. Bandwidth Interpretation

Modern NAND-based SSDs can provide high sequential data-transfer rates, but high sequential throughput does not mean Flash has DRAM-like random-access latency. DRAM is much more appropriate for fine-grained memory operations required during instruction execution and data processing.

6. Hierarchical Placement

A practical hierarchy is CPU registers/cache → DRAM → NAND Flash. When an application is launched, its persistent data can be read from NAND storage into DRAM. Once in DRAM, the processor can access the active working set efficiently. This separation allows Flash to provide capacity and persistence while DRAM provides speed.

7. Critical Thinking

Using NAND Flash as a direct replacement for DRAM would reduce system responsiveness because active computations would encounter much higher access latency. Conversely, using DRAM as permanent storage would require continuous power and would lose information during power failure. Therefore, the two technologies serve different levels of the hierarchy.

8. Evidence and Validation

- Measure sequential and random read/write throughput.
- Measure access latency for small and large transfers.
- Measure application startup time from Flash to DRAM.
- Measure DRAM bandwidth under application workloads.
- Compare energy and storage capacity for persistent data.

9. Conclusion

DRAM is appropriate for fast, volatile working memory, while NAND Flash is appropriate for persistent, high-density storage. The technologies complement each other in a hierarchical memory system.

5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Compare MRAM and RRAM

1. Scope of Items

The assessment asks for comparison of MRAM and RRAM based on access time, throughput and energy efficiency. A deeper comparison also considers non-volatility, endurance, density, write energy, retention and application suitability. [filecite turn1 file0 L70-L70](#)

2. Detailed Comparison

Parameter	MRAM	RRAM
Storage principle	Magnetic state	Resistance state
Volatility	Non-volatile	Non-volatile
Access time	Fast in suitable implementations	Potentially fast
Throughput	High potential	High potential
Refresh	Not required	Not required
Standby energy	Low due to non-volatility	Potentially low
Endurance	Generally high for suitable MRAM types	Technology-dependent
Density	Technology-dependent	Potentially high
Write energy	Implementation-dependent	Implementation-dependent
Potential applications	Embedded/persistent memory	Dense NVM/emerging computing

3. MRAM Technology

MRAM stores information using magnetic states rather than electrical charge in a conventional volatile cell. Its non-volatility allows information to remain stored without refresh. This characteristic can be useful for embedded systems that need rapid startup and persistent information.

4. RRAM Technology

RRAM stores information by changing the resistance of a material or device structure. It is also non-volatile and is being investigated for dense memory and emerging computing architectures. Its performance depends strongly on the specific RRAM technology and device implementation.

5. Access Time and Throughput

Both technologies have the potential for fast access compared with conventional persistent storage. However, actual access time and throughput depend on device architecture, controller, memory array design and manufacturing technology. Therefore, a rigorous comparison should use measured device data rather than claiming a single universal performance number.

6. Energy Efficiency

The absence of refresh gives both technologies an important potential standby-energy advantage. MRAM can provide efficient persistent access, while RRAM may provide high density and low-power potential. Write energy must be considered separately because programming a non-volatile memory cell can consume more energy than a read operation.

7. Critical Thinking

MRAM may be a stronger choice when endurance, fast persistent access and embedded reliability are major priorities. RRAM may be attractive when density, low-power operation or compute-in-memory concepts are important. The final selection should consider access time, throughput, endurance, retention, energy, density, process integration and cost.

8. Evidence and Validation

- Measure read and write latency separately.
- Measure random and sequential throughput.
- Measure energy per read and write operation.
- Measure standby power and retention.
- Evaluate endurance through repeated programming cycles.
- Compare error rates and reliability under operating conditions.

9. Conclusion

MRAM and RRAM are both promising non-volatile memory technologies. MRAM is attractive for fast and durable persistent memory applications, while RRAM offers opportunities in high-density and emerging low-power memory architectures. The correct choice depends on the measured requirements of the target system.